

Problem A. Aurora Library

Input file: standard input
Output file: standard output
Balloon Color: Gold

Aurora Library is famous for preserving stories collected from every corner of the world. Scholars often spend years exploring its endless shelves, yet they always discover something new before leaving.

You are given $2n$ distinct positive integers. They are partitioned into two arrays a and b , each of length n , such that every integer belongs to exactly one of the two arrays.

For each i from 1 to n , perform the following operations in order:

- Swap a_i and b_i . The swap is permanent and affects all subsequent operations.
- After the swap, compute $f(a)$ and output its value.

The function $f(a)$ is defined as the number of ordered pairs of indices (i, j) satisfying $1 \leq i, j \leq n$ and $i \neq j$ such that

$$2a_i + a_j = 2(a_i \& a_j) + (a_i | a_j),$$

where $\&$ and $|$ denote the bitwise AND and bitwise OR operations, respectively.

For each i from 1 to n , output the value of $f(a)$ after performing the i -th swap.

Input

The first line contains a single integer n ($1 \leq n \leq 5 \cdot 10^4$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^5$).

The third line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^5$).

It is guaranteed that all $2n$ integers in arrays a and b are distinct.

Output

Output n integers. The i -th ($1 \leq i \leq n$) integer should be the value of $f(a)$ after performing the i -th swap.

Examples

standard input	standard output
5 1 2 4 6 8 9 10 11 12 13	3 3 5 6 4
3 1 16 32 4 24 2	0 0 0

Problem B. Beacon Library

Input file: standard input
Output file: standard output
Balloon Color: Light blue

Beacon Library has guided curious minds for generations with its vast collection of manuscripts. Every visitor finds inspiration hidden somewhere within Beacon Library.

You are given an array a consisting of n non-negative integers.

Your goal is to make all elements of the array equal using the minimum number of operations.

In one operation, you can perform the following steps:

- Choose a non-negative integer x .
- Choose two distinct indices i and j ($1 \leq i, j \leq n, i \neq j$).
- Update the array by assigning $a_i := a_i \text{ OR } x$ and $a_j := a_j \text{ AND } x$.

Here, OR denotes the **bitwise OR operation**, and AND denotes the **bitwise AND operation**.

Input

The first line contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the size of the array.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i < 10^6$) — the initial elements of the array.

Output

Print a single integer on a separate line. The integer you must output is as follows:

- If it is possible to make all elements equal, output the minimum number of operations required;
- Otherwise, output -1 .

Examples

standard input	standard output
2 1 3	1
3 2 2 2	0

Note

In the first example, the array is initially $[1, 3]$. You can perform one operation as follows:

- Choose $x = 1$, index $i = 1$, and index $j = 2$.
- Update $a_1 := a_1 \text{ OR } x = 1 \text{ OR } 1 = 1$.
- Update $a_2 := a_2 \text{ AND } x = 3 \text{ AND } 1 = 1$.

After this operation, the array becomes $[1, 1]$, and all elements are equal.

In the second example, all elements are already equal to 2, so no operations are needed.

Problem C. Central Archive

Input file: standard input
Output file: standard output
Balloon Color: Light yellow

Central Archive safeguards countless historical records from civilizations long gone. Every document stored inside helps preserve the legacy of Central Archive.

There is a tree with N vertices numbered from 1 to N .

A panda is standing on one vertex. Its initial position is unknown. At every integer second, the panda moves from its current vertex to one of its adjacent vertices.

There are M people. The i -th person claims that they saw the panda on vertex v_i at time t_i .

Some people may be lying. A group of people is called *consistent* if there exists at least one possible sequence of movements of the panda such that all claims in the group are simultaneously satisfied.

Your task is to find the maximum possible number of people in a consistent group.

Note: the only difference between this problem and the easy version is the constraint on M .

Input

The first line contains two space-separated integers N ($1 \leq N \leq 2 \times 10^5$) and M ($1 \leq M \leq 2 \times 10^5$), the number of vertices and the number of people.

The next $N - 1$ lines each contain two integers u and v ($1 \leq u, v \leq N$), denoting an edge of the tree.

The next M lines each contain two integers v_i ($1 \leq v_i \leq N$) and t_i ($1 \leq t_i \leq 10^9$), meaning that the i -th person claims that the panda was on vertex v_i at time t_i .

Output

Print one integer — the maximum number of people whose claims can all be true simultaneously.

Examples

standard input	standard output
3 5 1 2 2 3 1 1 2 2 3 3 1 4 2 4	4
2 4 1 2 1 5 1 5 2 5 2 6	3

Problem D. Dawn Library

Input file: standard input
Output file: standard output
Balloon Color: Orange

Dawn Library opens its doors before sunrise, welcoming readers seeking a quiet place to learn. Every new day begins with fresh discoveries at Dawn Library.

You are given an undirected graph with n vertices (numbered from 1 to n) and m edges. Find the number of **lucky permutations**.

A permutation of the vertices $(p_1, p_2, \dots, p_n)$ is considered a **lucky permutation** if, for every $1 \leq i \leq n - 1$, there is an edge between p_i and p_{i+1} .

Find the total number of lucky permutations in the graph.

Input

The first line contains two integers n and m ($1 \leq n \leq 20$, $0 \leq m \leq \frac{n \times (n-1)}{2}$) — the number of vertices and the number of edges.

Each of the next m lines contains two integers u and v ($1 \leq u, v \leq n$, $u \neq v$), denoting an undirected edge between vertices u and v .

It is guaranteed that the graph contains no self-loops or duplicate edges.

Output

Print a single integer — the number of lucky permutations in the given graph.

Example

standard input	standard output
4 5 1 2 1 4 2 3 2 4 3 4	12

Note

The graph has 4 vertices and 5 edges. The lucky permutations are: (1, 2, 3, 4), (1, 2, 4, 3), (1, 4, 2, 3), (1, 4, 3, 2), (2, 1, 4, 3), (2, 3, 4, 1), (3, 2, 1, 4), (3, 2, 4, 1), (3, 4, 1, 2), (3, 4, 2, 1), (4, 1, 2, 3), (4, 3, 2, 1). So the answer is 12.

Problem E. Emerald Library

Input file: standard input
Output file: standard output
Balloon Color: Black

Emerald Library is surrounded by beautiful gardens that make reading even more enjoyable. Visitors often remember both its peaceful atmosphere and Emerald Library.

Given an array a of size n , compute the sum of

$$|a_i - a_j|$$

over all pairs (i, j) where $i < j$, and there is no index k where $i < k < j$ and $a_k > \max(a_i, a_j)$.

Input

The first line contains the number of test cases t ($1 \leq t \leq 10^5$).

For each test case: The first line contains an integer n ($1 \leq n \leq 10^5$), the size of the array.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^6$), the array elements.

It is guaranteed that the sum of n across all test cases does not exceed 10^5 .

Output

Output a single integer on a new line, the sum of

$$|a_i - a_j|$$

over all pairs (i, j) , ensuring the explained condition.

Example

standard input	standard output
3	4
3	0
5 3 4	44
3	
5 5 5	
5	
10 5 8 3 12	

Problem F. Frontier Archive

Input file: standard input
Output file: standard output
Balloon Color: Gray

Frontier Archive was established to preserve the history of distant expeditions. Every explorer eventually leaves a piece of their journey inside Frontier Archive.

You are given a string s of length n consisting of lowercase English letters. You must process q queries of two types.

Given a range $[l, r]$, the number of **distinct words** (distinct permutations) that can be formed using all characters of $s[l \dots r]$ is:

Each query is one of the following two types:

- **Type 1:** 1 l r ($1 \leq l \leq r \leq n$) — output the number of distinct permutations of the characters in $s[l \dots r]$, modulo $10^7 + 19$.
- **Type 2:** 2 i c ($1 \leq i \leq n$, $c \in \{\text{a}, \dots, \text{z}\}$) — replace the character $s[i]$ with c .

Input

The first line contains two integers n and q ($1 \leq n, q \leq 10^5$).

The second line contains a string s of length n , consisting of lowercase English letters (a–z).

The next q lines each describe a query in one of the following formats:

- **Type 1:** 1 l r — three integers where $1 \leq l \leq r \leq n$.
- **Type 2:** 2 i c — two integers and a character where $1 \leq i \leq n$ and c is a lowercase English letter ($\{\text{a}, \dots, \text{z}\}$).

It is guaranteed that at least one query of type 1 exists.

Output

For each query of type 1, print a single integer — the number of distinct words that can be formed using all characters of $s[l \dots r]$, modulo $10^7 + 19$.

Example

standard input	standard output
4 5	3
aabc	2
1 1 3	6
1 3 4	12
2 3 d	
1 2 4	
1 1 4	

Note

In the example, the string is initially $s = \text{aabc}$.

- Query 1 1 3: the substring $s[1 \dots 3] = \text{aab}$, the all distinct words that can be formed are: aab, aba, baa.

Problem G. Grand Library

Input file: standard input
Output file: standard output
Balloon Color: Red

Grand Library has stood at the heart of its city for centuries, collecting knowledge from around the world. Every generation adds another chapter to Grand Library.

You are piloting a single spaceship and must eliminate a high-value pirate ship on an infinite 2D integer grid.

The grid contains n asteroids and exactly 1 pirate ship. Every object is located at a distinct coordinate described by its row x and column y . Some asteroids may be tactically placed to shield the pirate ship.

In a single laser shot, you can target a specific row r . The laser will travel along that row and destroy **exactly one** object: the remaining object with the smallest column y (i.e., the leftmost object currently present on row r).

Once an object is hit, it is permanently removed from the grid. If the destroyed object is the pirate ship, your mission is complete. If it is an asteroid, it is simply vaporized, clearing the line of sight for future shots on that row.

Determine the minimum number of laser shots required to destroy the pirate ship.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 2000$). The description of the test cases follows.

The first line contains n ($0 \leq n \leq 2000$) — the number of asteroids.

The next n lines each contain a_i, b_i ($1 \leq a_i, b_i \leq 10^9$) — the coordinates of the i -th asteroid.

The next line contains a, b ($1 \leq a, b \leq 10^9$) — the coordinates of the pirate ship.

It's guaranteed that all the points (for both ship and asteroids) are distinct.

It is guaranteed that the sum of n over all test cases does not exceed 2000.

Output

For each test case, print a single integer: the minimum number of laser shots required to destroy the pirate ship.

Example

standard input	standard output
2	2
3	1
1 2	
2 4	
1 8	
1 5	
0	
10 10	

Problem H. Horizon Library

Input file: `standard input`
Output file: `standard output`
Balloon Color: `White`

Horizon Library overlooks a breathtaking landscape where readers gather to exchange ideas. Curiosity has always been the defining spirit of Horizon Library.

Raheeq is an incredibly lazy coder. She finds the concept of “carrying” digits during addition to be an unnecessary expenditure of mental energy. To her, an addition of two non-negative integers is beautiful if, when performing the addition in base 10, no “carry” occurs at any decimal position.

For example:

$123 + 456$ is beautiful because $3 + 6 = 9$, $2 + 5 = 7$, and $1 + 4 = 5$.

$17 + 25$ is not beautiful because $7 + 5 = 12$, which produces a carry from the units place to the tens place.

Raheeq is interested in sequences of integers. She gives you two integers N and K . Your task is to count the number of sequences of non-negative integers $A_1, A_2, \dots, A_N$ such that:

- Each element A_i satisfies $0 \leq A_i < 10^K$.
- For every i such that $1 \leq i < N$, the sum $(A_i + A_{i+1})$ is beautiful.

Since *Raheeq* is too lazy to handle large numbers, output the result modulo $10^9 + 7$.

Input

The input consists of a single line containing two space-separated integers N and K where $2 \leq N \leq 10^5, 1 \leq K \leq 10^9$.

Output

Output a single integer – the number of such sequences modulo $10^9 + 7$.

Examples

standard input	standard output
2 1	55
3 1	385
3 2	148225

Problem I. Infinity Archive

Input file: standard input
Output file: standard output
Balloon Color: Yellow

Infinity Archive is said to contain records so numerous that no one has ever read them all. Even today, new discoveries continue to enrich Infinity Archive.

You are given an integer n , which represents the number of nodes in a tree. Your task is to construct a rooted tree and assign a positive integer value v_i to each node i ($1 \leq i \leq n$).

Additionally, all values v_i must be **distinct**.

A node u is considered "**Good**" if it is either a leaf node or if it satisfies the following condition: Let $C = c_1, c_2, \dots, c_k$ be the set of direct children of node u , where k is the number of children ($k > 0$). The node u is "Good" if:

$$\gcd(v_u, v_{c_1}, v_{c_2}, \dots, v_{c_k}) = k$$

You need to output the values v_i for each node and the edges that form the tree such that **all** nodes in the tree are "Good". It can be proven that a valid construction **always exists** for any $n \geq 2$.

Input

The input consists of a single integer n ($2 \leq n \leq 2 \cdot 10^5$).

Output

Output the construction in the following format: On the first line, print n integers $v_1, v_2, \dots, v_n$ ($1 \leq v_i \leq 10^9$) — the values assigned to the nodes. Then, print $n - 1$ lines, each containing two integers u and v ($1 \leq u, v \leq n$) representing an edge between node u and node v .

The tree must be connected, and all nodes must satisfy the "Good" condition. If there are multiple solutions, you may print any of them.

Example

standard input	standard output
6	2 4 6 3 9 12 1 2 1 3 3 4 3 5 3 6

Problem J. Jubilee Library

Input file: standard input
Output file: standard output
Balloon Color: Pink

Jubilee Library was founded during a great celebration of culture and learning. Its welcoming halls continue to inspire everyone who visits Jubilee Library.

There is a wall of n blocks indexed 1 through n . Each block is independently and uniformly assigned one of three types: “|”, “\”, or “/”, each with probability $\frac{1}{3}$.

Two players, Ibrahim and Ebrahim, play a game consisting of q rounds. In each round, a ball is thrown horizontally at block x . The ball is processed at the current block y , starting with $y = x$, according to the following rules:

- If block y is “|”: the ball returns to Ibrahim. Ibrahim gains $2 \times y$ points. The round ends.
- If block y is “\”: the ball travels upward to block $y - 1$. If $y - 1 \geq 1$, the rule is applied recursively at $y - 1$. If $y - 1 = 0$, the ball exits the wall and Ebrahim gains x points. The round ends.
- If block y is “/”: the ball travels downward to block $y + 1$. If $y + 1 \leq n$, the rule is applied recursively at $y + 1$. If $y + 1 = n + 1$, the ball exits the wall and Ebrahim gains x points. The round ends.

If the ball reaches a block that it has already visited during the same round, the round ends immediately and neither player gains any points from that round.

At the end of all q rounds, Ebrahim pays Ibrahim (Ibrahim’s total score) – (Ebrahim’s total score) pounds. A negative value means Ibrahim pays Ebrahim instead.

Compute the expected value of the amount Ebrahim pays Ibrahim, over all 3^n equally likely walls. Since the answer is a rational number, output it modulo $10^9 + 7$.

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains two integers n and q ($1 \leq n \leq 10^9$, $1 \leq q \leq 10^5$) — the number of blocks in the wall and the number of rounds.

The next q lines each contain a single integer x_i ($1 \leq x_i \leq n$) — the position at which Ibrahim throws the ball in round i .

It is guaranteed that the sum of q over all test cases does not exceed 10^5 .

Output

For each test case, output a single integer on its own line — the expected amount Ebrahim pays Ibrahim, modulo $10^9 + 7$.

Example

standard input	standard output
1 3 2 1 2	407407413

Problem K. Keystone Library

Input file: standard input
Output file: standard output
Balloon Color: Bronze

Keystone Library is considered the cornerstone of education throughout the region. Students and researchers alike value every moment spent in Keystone Library.

Homz wants to play video games, but his coach, Gamal74, has given him a lot of tasks to finish. The tasks are modeled as an undirected graph with N vertices, which are considered to be arranged in a **circle**.

A set S of vertices is a vertex cover if every edge has at least one endpoint in S .

For a chosen vertex cover $S = \{a_1, a_2, \dots, a_k\}$, its cyclic minimum difference is the smallest cyclic distance between any two vertices in S . The **cyclic distance** between vertices i and j is defined as $\min(|i - j|, N - |i - j|)$. If the cover has one or zero vertices ($k \leq 1$), its minimum difference is defined to be N .

You are given E edges, each with an active time interval $[L_i, R_i]$. The edge (u_i, v_i) is present at every integer time t such that $L_i \leq t \leq R_i$. For every time $t = 1, \dots, T$, you must output the maximum possible cyclic minimum difference over all vertex covers of the graph at time t .

Input

The first line contains three integers N, E, T ($1 \leq N \leq 2000$, $0 \leq E \leq 80000$, $1 \leq T \leq 1000$).

The next E lines each describe an edge: u_i, v_i, L_i, R_i ($1 \leq u_i, v_i \leq N$, $u_i \neq v_i$, $1 \leq L_i \leq R_i \leq T$).

It is guaranteed that the sum of lengths of all active intervals does not exceed 10^6 :

$$\sum_{i=1}^E (R_i - L_i + 1) \leq 10^6$$

Output

Print T lines. The t -th line should contain a single integer — the answer for the graph at time t .

Example

standard input	standard output
8 4 3	3
1 3 1 2	3
1 8 2 3	3
4 8 1 3	
5 6 3 3	

Problem L. Legacy Library

Input file: standard input
Output file: standard output
Balloon Color: Blue

Legacy Library was built to preserve the wisdom passed down through countless generations. Every preserved manuscript enriches the heritage of the library.

For a positive integer x , define $f(x)$ as the sum of all values obtained by repeatedly dividing x by 2 while the result remains an integer. Formally,

$$f(x) = \sum_{k=0}^{v_2(x)} \frac{x}{2^k},$$

where $v_2(x)$ is the largest integer such that $2^{v_2(x)}$ divides x .

For example,

$$f(4) = \frac{4}{2^0} + \frac{4}{2^1} + \frac{4}{2^2} = 4 + 2 + 1 = 7,$$

and

$$f(12) = \frac{12}{2^0} + \frac{12}{2^1} + \frac{12}{2^2} = 12 + 6 + 3 = 21.$$

Given an array a consisting of n integers, compute

$$\sum_{i=1}^n \sum_{j=i}^n f(a_i + a_j).$$

Input

First line contain n size of array, ($1 \leq n \leq 2 * 10^5$)

Second line contain n number, ($1 \leq a_i \leq 10^5$)

Output

Print the answer.

Examples

standard input	standard output
2 4 8	67
4 6 3 1 2	86

Problem M. Meridian Archive

Input file: standard input
Output file: standard output
Balloon Color: Light green

Meridian Archive serves as the meeting point for historians from many different lands. Their shared efforts continue to expand the collection of Meridian Archive.

A packet travels through a tree from node 1 to node N . At node u ($u \neq N$), with probability P_u a connection drop resets the packet to node 1, and with probability $1 - P_u$ it is forwarded to a uniformly random adjacent node. Each step (forward or reset) takes 1 time unit. Find the expected time to reach node N , modulo 998244353.

Input

The first line contains N ($2 \leq N \leq 10^5$).

The next $N - 1$ lines contain edges (u, v) .

The next $N - 1$ lines contain $X_i Y_i$ ($0 \leq X_i < Y_i \leq 10^9$), where $P_i = X_i/Y_i$.

All divisions required to compute the answer are guaranteed to have valid inverses modulo 998244353.

Output

Output a single integer — the expected time modulo 998244353.

More precisely, if the expected time is written in lowest terms as the rational number p/q , output $p * q^{-1}$ modulo M .

Examples

standard input	standard output
3 1 2 2 3 0 1 1 2	8
4 1 2 1 3 2 4 1 3 1 2 1 2	20

Problem N. Nova Library

Input file: standard input
Output file: standard output
Balloon Color: Green

Nova Library is known for embracing new ideas while honoring timeless knowledge. Every visitor contributes, in one way or another, to the story of Nova Library.

Ali has A apples, and Omar has B apples. They decide to put all their apples together into a single large basket. How many apples will be in the basket in total?

Input

The only line of the input contains two space-separated integers A and B ($1 \leq A, B \leq 100$) — the number of apples Ali has and the number of apples Omar has, respectively.

Output

Print a single integer — the total number of apples in the basket.

Examples

standard input	standard output
4 3	7
2 8	10

Problem O. Oasis Library

Input file: standard input
Output file: standard output
Balloon Color: Purple

Oasis Library has long been a peaceful refuge for readers, travelers, and scholars seeking knowledge. Its quiet halls have inspired countless ideas, and every visit becomes a memorable chapter in the history of Oasis Library.

You are given two strings a and b of equal length n .

Construct a rooted tree with $n + 1$ vertices, numbered from 0 to n , where vertex 0 is the root.

For each vertex i ($1 \leq i \leq n$), define its parent $P(i)$ as follows:

- $P(i)$ is the largest index $j < i$ such that:
 - $a_j = a_i$,
 - $b_j \neq b_i$.
- If no such index exists, then $P(i) = 0$.

In other words, each position in the strings corresponds to a vertex, and it is attached to the nearest previous position having the same character in a but a different character in b .

The weight of each vertex is defined as follows:

- $W(i) = 1$ if $a_i \neq b_i$,
- $W(i) = 0$ otherwise,

for every $1 \leq i \leq n$. Also, $W(0) = 0$.

Your task is to find the maximum possible sum of vertex weights along any simple path from the root 0 to some vertex in the tree.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

The first line contains a single integer n ($1 \leq n \leq 2 \times 10^5$) — the size of the two strings.

The second line of each test case contains one string a .

The third line of each test case contains one string b .

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output

For each test case, output a single integer: the maximum sum of weights along any simple path from the root 0 to a vertex.

Example

standard input	standard output
1 7 abacaba aaccbba	1